

Exercise series 4:

Rel. Algebra To SQL

Introduction to databases



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Advanced Database Research & Modelling
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Problem 1: Dealers, Producers and Products

Consider the example data provided in the tables Dealers, Producers, Products, and Deals shown below.

- Relax: <https://dbis-uibk.github.io/relax/calc/gist/e0509d68845aaefacf3d36500ed56ccc>
- DB_fiddle: <https://www.db-fiddle.com/f/itidCoYCpBTsgf1jsRPEkd/0>

Dealers	
dealer_id	name
1	Bob's store
2	Appliances Inc.
3	Quality Goods
4	Tech Market

Producers	
producer_id	name
1	United Technics
2	Technics Inc.
3	Big Bang
4	ACME

Products			
product_id	type	price	producer_id
1	vacuum cleaner	66.9	3
2	washing machine	399.99	2
3	tumble dryer	500	2
4	washing machine	780	3
5	oven	175.45	1
6	washing machine	545.9	1
7	vacuum cleaner	475.19	4
8	washing machine	519.99	4
9	tumble dryer	600	2

Deals	
producer_id	dealer_id
1	1
1	2
2	2
2	4
3	1
3	2
3	3
4	2
4	3

You may assume that a dealer, who has a deal with a given producer, sells all the products produced by that producer.



Problem 1: Dealers, Producers and Products

- 1) Give all types of products that are produced by at least two different producers.
- 2) Give the minimum price of a product sold by Quality Goods.
- 3) Give the name of the producer of the most expensive washing machine.
- 4) Give the names of the dealers that do not sell any vacuum cleaners.



Problem 2: Employees and Departments

Consider the example data provided in tables Employees and Departments below.

- Relax: <https://dbis-uibk.github.io/relax/calc/gist/1c947ca70bbc6efffae62eb19d16d023>
- DB_fiddle: <https://www.db-fiddle.com/f/itidCoYCpBTsgf1jsRPEkd/1>

Employees		
name	salary	d_name
John	10000	R & D
John	8000	ICT
Paul	12000	HR
George	8000	ICT
George	7000	R & D
Susan	11000	Fin.
Susan	7000	ICT

Departments	
name	manager
R & D	Bill
HR	Bob
ICT	Bill
PR	Helen
Fin.	Alan
Law	Bob

Note that an employee can be assigned to multiple departments, and a salary is specific to the work done in that department.



Problem 2: Employees and Departments

- 1) Give all departments in which all employees have different salaries.
- 2) Give all pairs of employees that have the same salary in the same department. Make sure you give every pair just once.
- 3) Give all employees assigned to all departments in which Bill is manager.
- 4) Give all employees assigned to any department in which Bob is manager.



Problem 3: Exam question (in 2017)

Consider the example data provided in the following tables:

- Relax: <https://dbis-uibk.github.io/relax/calc/gist/c42da7b7855e525e80e0a7255568e587>
- DB_fiddle: <https://www.db-fiddle.com/f/itidCoYCpBTsgf1jsRPEkd/4>

Assignment	
project_id	emp_id
1001	5
1001	7
1002	9
1002	2
1003	3
1004	3
1004	8
1005	1
1006	6

Department	
dept_id	name
1	Marketing
2	Finance
3	Human Resources
4	R&D
5	Management

Project	
project_id	name
1001	New Training
1002	Bus Stop Billboards
1003	Recruitment Drive
1004	Brain Drain
1005	The Ultimate Product
1006	Solo Sing Career

Employee				
emp_id	name	sex	salary	dept_id
1	Bob	M	26,000	4
2	Bill	M	49,000	5
3	Jim	M	28,000	3
4	Jill	F	48,000	5
5	Jane	F	35,000	1
6	Hank	M	35,000	4
7	Joan	F	37,000	2
8	Tom	M	39,000	4
9	Ron	M	29,000	1

Note that an employee can be assigned to multiple departments, and a salary is specific to the work done in that department.



Problem 3: Exam question (in 2017)

- 1) Give the names of all employees currently working on more than one project.
- 2) Give the name of the highest paid employee working on Brain Drain.
- 3) Give the names of all projects on which only men are working.
- 4) Give the names of all departments that currently have no employees involved in any projects.
- 5) Give the names of departments where all employees are active in a project.

